

Fraud-Resistant Meal Voucher Redemption and Real-Time Operational Analytics for Conference Management: Design and Evaluation of ConferenceCheck Mobile

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Repository: <https://github.com/ndashi-chisanga6/ConferenceCheck-Mobile-Analytics-Meals-and-Notifications> Validated state: tag milestone-12-final-submission

Declaration

I declare that this report is my own work. It has not been submitted, in whole or in part, for any other award at this or any other institution, and every source I have drawn on is acknowledged in the reference list.

The use of artificial intelligence tools during this project is disclosed in **Appendix A**, which records what each category of tool was used for, what was provided to it, and the limits placed on its use by Section 9 of the approved project proposal. All evaluation data in this study is seeded and synthetic: attendees, meal vouchers and sessions in the simulated-event exercise are generated by `tools/seed-evaluation-event.php` and describe no real conference or real person. No respondent or attendee data belonging to any third party was uploaded to any external tool at any point.

Signed: Date:

Conventions used in this report

Latencies are quoted in milliseconds to one decimal place in tables and rounded to the nearest whole millisecond in prose. Counts and rejection rates are exact throughout. Timings are wall-clock seconds or milliseconds for a single named run on PHP 8.5.8 with PostgreSQL 18, and the run is always named. Where a number also appears in `docs/validation_report.md`, the two

are the same measurement and that file is the reproducible source; this report does not re-derive figures independently of it.

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Abstract

On-site conference operations, meal service, session seating and attendee communication, are still widely managed with paper registers and printed vouchers, processes that scale poorly and are vulnerable to duplicate-collection fraud. This report presents the design, implementation and evaluation of ConferenceCheck Mobile: Analytics, Meals and Notifications, component 4.3b of a two-part conference management companion application whose sibling component, 4.3a, delivers attendee check-in capture and offline synchronisation separately. This project consumes check-in data through a shared REST API contract and contributes four capabilities that carry equal weight in its evaluation: a QR-code based meal voucher protocol in which single-use redemption is enforced structurally, through database-level row locking and an append-only redemption relation with a per-voucher uniqueness constraint, rather than through application-level checks; a near-real-time analytics dashboard with a bounded 30-second staleness guarantee; session attendance tracking with configurable capacity thresholds and automatic overcrowding alerts; and dual-channel attendee notifications, combining Firebase Cloud Messaging push with an authenticated in-app feed carrying per-recipient delivery records. An offline scan queue, whose safe replay follows from the server's idempotent scan semantics, extends the design to intermittent venue connectivity. The system was evaluated against a seeded 500-attendee event with three meal categories and a voucher per attendee per category, ten sessions and scripted concurrent-fraud scenarios, driven through the live API rather than through unit-level mocks. Every criterion in the proposal's Table 3 was met: 100% duplicate-redemption rejection across 100 sequential and 20 concurrent duplicate

submissions, with every concurrency race resolving to exactly one success; all three categories redeemable independently for every one of twenty attendees holding a voucher in each; median voucher-scan latency of 187.5 ms (95th percentile 286.7 ms); dashboard staleness bounded at approximately 30.2 seconds; and byte-exact CSV exports against database ground truth for both attendance and meal redemptions. A live, human-staffed manual baseline was not run, and this is stated as a limitation rather than approximated with a fabricated figure; the qualitative argument instead rests on the system's measured 188 ms scan-to-decision time and the fact that a paper voucher has no server-side check to fail. The results indicate that placing anti-fraud guarantees in the database's constraint machinery, rather than in application code, eliminates time-of-check-to-time-of-use races by construction while keeping scan-point latency well within interactive bounds.

1. Introduction

Modern conferences concentrate hundreds of attendees, multiple parallel sessions and time-boxed catering windows into a short operational period. The quality of the attendee experience depends on how efficiently organisers can verify entitlement, observe live operational state, and communicate changes as they happen. Traditional approaches rely on fragmented, largely manual tools: printed meal tickets, paper attendance registers and public-address announcements. These methods have three structural weaknesses. Paper vouchers are trivially duplicated or reused, and manual meal lines create congestion at service points. Organisers have no consolidated, real-time view of attendance or consumption, so operational decisions are made on stale or anecdotal information. Attendee communication is broadcast-only and unverifiable, with no record of who received a given update.

Quick Response (QR) codes provide a mature, standardised mechanism for machine-readable identification [1], and QR-based attendance has been proposed as a way to reclaim the lecture time that manual registers consume [3]. Cloud push-messaging services such as Firebase Cloud Messaging exist to send messages to mobile devices reliably [9], within the delivery guarantees Section 2 sets out, and modern cross-platform toolkits such as Flutter allow a single codebase to deliver the required scanning, charting and notification functionality [19]. This project applies these technologies to conference operations as component 4.3b of the ConferenceCheck capstone programme. A sibling component, 4.3a, handles the core attendee check-in capture and offline synchronisation engine separately; this project does not rebuild those features, and instead consumes check-in data through a shared REST API contract and focuses on what happens after check-in.

1.1 Problem statement

Conference organisers lack integrated, trustworthy tools for the operational tasks that follow attendee registration and check-in. Meal distribution based on printed vouchers or name lists cannot enforce single use; duplicate collection is difficult to detect at the point of service, and reconciliation after the event is manual and unreliable. Attendance and check-in data, even when captured digitally, is not surfaced to organisers in real time, so decisions about catering

quantities, room allocation and staffing are made on guesswork. Session rooms have physical capacity limits, but organisers have no mechanism that tracks per-session attendance as it happens or warns them before rooms overcrowd. Communication with attendees is one-directional and unlogged, with no record of which attendees were reachable for a given schedule change. The consequence is operational inefficiency, measurable fraud exposure in catering budgets, safety risk from overcrowded rooms, and a degraded attendee experience.

1.2 Aim

To design, implement and evaluate the analytics, meal management, session capacity, notification and reporting component of the ConferenceCheck Mobile application, providing conference organisers with real-time operational insight, fraud-resistant meal voucher redemption and auditable attendee communication.

1.3 Objectives

1. To design and implement a real-time analytics dashboard showing check-in progress, check-in rates and trends for event organisers.
2. To build a QR-code based meal voucher scanning and redemption system that enforces single-use redemption across multiple meal categories.
3. To implement session attendance tracking with capacity management and overcrowding alerts.
4. To build a push notification system for attendee communications and schedule updates, with per-recipient delivery records.
5. To implement reporting and data export (CSV) functionality for event organisers.
6. To produce comprehensive technical documentation and a publication-ready paper.

1.4 Scope

The project covers Flutter mobile modules for the analytics dashboard, meal voucher generation and scanning, session attendance and capacity alerts, a notification inbox and report access, together with the corresponding server-side business logic on the CMS v4 Laravel backend. It integrates with component 4.3a's core check-in module through a shared REST API contract, consuming its check-in data as a read-only input to analytics; check-in scanning, attendee search and the offline synchronisation engine are component 4.3a's remit and are out of scope here. Evaluation is performed on seeded, realistic event datasets including concurrent-scan fraud scenarios, described in full in Section 5. Payment processing, external ticketing integration and a full web-based administration panel are out of scope for both components.

1.5 Contributions

This work makes four contributions, each carrying equal weight in the evaluation that follows. The first is a redemption protocol in which the single-use guarantee is carried by the database's constraint machinery: a unique constraint on an append-only redemption relation, enforced inside a transaction with a row lock on the voucher, rather than by an application-level check,

which eliminates time-of-check-to-time-of-use races by construction rather than by defensive coding. The second is a near-real-time analytics dashboard whose staleness bound is a stated design parameter (a 30-second poll) rather than an incidental property, chosen explicitly over introducing a WebSocket layer within the project's scope. The third is session capacity tracking whose overcrowding alerts are dispatched automatically on a threshold transition, converting a safety problem into an alert delivered before a room reaches capacity rather than a discovery made after. The fourth is a dual-channel notification design, push plus an authenticated in-app inbox with per-recipient delivery records, that treats Firebase Cloud Messaging's best-effort delivery model as a known property to be compensated for rather than assumed away, together with an offline scan queue whose replay safety is a direct corollary of the same idempotent server design that makes the redemption protocol safe.

2. Literature Review

The QR Code symbology is standardised in ISO/IEC 18004 [1], which specifies its data capacity and, critically for verification applications, its four selectable error-correction levels, allowing codes to remain machine-readable when partially damaged or poorly displayed on low-brightness screens. Tiwari [2] surveys the structural properties that drove QR adoption over one-dimensional barcodes in ticketing and access-control contexts: high capacity in a small print area, omnidirectional scanning, and decode robustness on commodity camera hardware. A key design implication, adopted in this project, is that the QR payload should be an opaque server-issued token rather than semantically meaningful data, so that possession of the printed code confers no information and validation authority remains entirely server-side.

Masalha and Hirzallah [3] propose a QR-based attendance system for a university setting, motivated by the observation that taking attendance manually consumes around ten minutes of each lecture, and they project that their design would remove most of that cost. Their paper describes the design and analyses it; it reports no experiment, and in particular no measured comparison against a manual register, so it should not be read as evidence of how much time or error such a system saves in practice. Its analysis does, however, identify the central weakness of naive QR attendance directly: a student outside the room can be sent an image of the displayed code by a student inside it, so any anti-fraud property must come from what the server does with a scan, not from the code itself. Their own answer to that is to bind the scan to a place and a person, through device location and a facial match. That weakness, rather than their proposed remedy for it, is what motivates the server-side single-use redemption design evaluated in this report, in which the database, not the scanning client, is the arbiter of whether a voucher has already been consumed.

That weakness has not gone away in the decade since, and the recent literature answers it by constraining the scan rather than the entitlement. Mohammed and Zidan [4] regenerate the code twice a second, so that a code which has been photographed is already stale by the time it is presented again; they report improved accuracy and a positive reception from students and instructors, drawn from surveys and record analysis rather than from a controlled comparison. Nwabuwe et al. [5], writing in the same journal as Masalha and Hirzallah nine years later, still

describe a shared code as the central vulnerability, and bind the scan to a place and a device: the venue's polygonal coordinates are encoded in the code itself and checked by geofence, and the registering handset's IMEI is verified so that one person cannot register for another. Alsuhibany [6] secures the symbol instead of the scan, embedding a watermark so that a substituted or tampered code can be detected, and reports high accuracy over a dataset of 5,000 samples.

Each of these binds the act of scanning more tightly, to the moment, the place, the device or the integrity of the symbol, and each is a reasonable answer to the problem it takes on. None of them makes the thing being claimed single-use, which is a different guarantee and the one this project pursues. The distinction is in what is being recorded: an attendance mark may legitimately be registered once per person per session, and the fraud is someone registering who is not there, whereas a meal entitlement is consumed, and the fraud is the same person consuming it twice. Binding a scan to a place or a handset does not prevent the second of these, because the attendee is genuinely present and holding their own phone both times.

It is also worth recording what this body of work does not contain. None of [4], [5] or [6] reports a measured comparison against a manual process, and neither does [3]. Section 6.8 therefore has no external figure to set its own against, and says so rather than implying that one exists.

For the redemption transaction itself, Helland [16] provides the canonical treatment of idempotence in distributed systems: operations that may be retried, as scans on unreliable venue networks will be, must be designed so that repeated submission cannot produce repeated effect. In this project, that principle is realised by recording redemptions in an append-only table with a database-level uniqueness constraint on the voucher identifier, making double-redemption a constraint violation rather than an application-level race condition, and it is the same property that makes the offline scan queue described in Section 3.5 safe to replay without reconciliation logic.

On the presentation side, Few [7] establishes design principles for operational dashboards: a dashboard must fit on a single screen, prioritise the small number of measures that drive action, and use visual encoding rather than tables of numbers for trend perception. Nielsen [8] provides the response-time thresholds that inform this project's latency targets: feedback within one second preserves a user's flow of thought, which sets the budget for scan-validation round trips at a catering service point where queue throughput matters, and is the threshold against which Section 6.4 reports the measured results.

For attendee communication, Firebase Cloud Messaging documentation describes an architecture in which each app instance registers for messages and is identified by its own registration token [10], and states that a message accepted by FCM has not thereby been delivered: a message for an offline device is stored and delivered when the device reconnects, and delivery may be delayed or may not happen at all [11]. Delivery is therefore best-effort rather than guaranteed, tokens expire, and devices without Google Play services are unreachable, all of which motivate the dual-channel design evaluated in Section 6.7, in which every notification is also persisted to an in-app inbox with per-recipient delivery records. The

service interface between the mobile client and backend follows the REST architectural style [12]; endpoint authentication uses bearer tokens as specified in RFC 6750 [13] and implemented by Laravel Sanctum [14], with mobile-specific hardening guided by the OWASP Mobile Application Security Verification Standard [15]. The client architecture applies the separation of presentation, state and data access descended from the Model-View- Controller pattern [17], realised in Flutter's widget and provider model [19].

Taken together, this body of work indicates that the individual technologies are mature, but that existing systems apply them piecemeal: attendance tools do not manage catering, catering tools do not track session capacity, and none provide the organiser with a consolidated live operational view. This project's contribution, set out in Section 1.5, is that integration, with each design decision traceable to a specific finding in this literature rather than to convention.

3. Methodology

3.1 System architecture and the component 4.3a / 4.3b boundary

The system is a client-server architecture with three tiers: a Flutter mobile application, the CMS v4 Laravel backend exposing a REST API [12], and a PostgreSQL database, with Firebase Cloud Messaging as the external push channel.

Figure 1. System architecture and the component 4.3a / 4.3b boundary.

ConferenceCheck Mobile: Flutter application (Android)	
Component 4.3a: Core Check-in and Sync Engine (<i>sibling, developed separately, out of scope</i>): attendee check-in capture; offline synchronisation engine	Component 4.3b: Analytics, Meals and Notifications (<i>this project</i>): analytics dashboard; meal voucher scan and redemption; session capacity and overcrowding alerts; notification inbox; reports and CSV export
Shared REST API contract, Sanctum bearer-token authenticated. 4.3b consumes 4.3a's check-in data as seeded fixtures during independent development	
CMS v4 Laravel REST API + PostgreSQL database (shared backend)	
Push dispatch: Laravel → Firebase Cloud Messaging (external) then to the device, for component 4.3b notifications only	

The mobile application follows a layered structure derived from the Model-View-Controller separation [17]: a presentation layer of Flutter widgets, a state layer of Riverpod providers holding screen state and orchestrating asynchronous calls, and a data layer of repository classes wrapping a single authenticated REST client. Platform services, camera access via `mobile_scanner` and push token registration via `firebase_messaging`, are isolated behind interfaces so that they can be stubbed in widget tests. The backend extends the CMS v4 Laravel application. Every endpoint requires a Sanctum-issued bearer token [13, 14] and returns a uniform JSON envelope with success, message and data fields. Authorisation is role-based: organiser accounts manage events and view analytics, scanner accounts may only submit scan validations, and attendee accounts may only read their own vouchers and notifications.

3.2 Data model and integrity guarantees

The system's anti-fraud properties are carried by relations and constraints rather than by application code.

Table 1. Core relations and the integrity constraints underpinning system guarantees.

Relation	Purpose	Key integrity constraint
attendees	Event attendee records, shared with 4.3a	Unique ticket_code; unique qr_token
meal_categories	Meal windows, for example Breakfast, Lunch, VIP Lunch	Owned by event; time-windowed
meal_vouchers	One voucher per attendee per category	Unique qr_token; unique (attendee, category) pair
meal_redemptions	Immutable, append-only record of each redemption	Unique meal_voucher_id enforces single use
event_sessions / session_attendance	Conference sessions and per-session attendance	Unique (session, attendee) prevents duplicate attendance
device_tokens / notification_recipients	FCM registration and per-recipient delivery records	Delivery auditable per user and per notification

3.3 The voucher redemption protocol

Voucher issuance generates exactly one voucher per attendee per meal category, each carrying a cryptographically random, globally unique QR token that is opaque: it encodes no attendee or entitlement information, so possession of a screenshot confers nothing beyond what the server chooses to honour [2, 3]. The protocol has four steps. A scanner device decodes the QR token and submits it with its own device identifier to the scan endpoint. The server resolves the token to a voucher, verifying that it belongs to the event being served and that the category's service window is open. Within a database transaction, the server locks the voucher row (`lockForUpdate()`) and inserts a redemption record; the unique constraint on the voucher identifier guarantees that if two scanners submit the same token concurrently, exactly one

insert succeeds and the other receives a definitive "already redeemed" response naming the earlier redemption time [16]. The redemption record, including the redeeming device identifier and timestamp, is immutable thereafter, providing a complete audit trail for post-event reconciliation. This design places the single-use guarantee in the database's constraint machinery rather than in application-level checks, eliminating time-of-check-to-time-of-use races by construction. The concurrency behaviour this protocol is designed to produce is measured directly in Section 6.2.

3.4 Session attendance and capacity alerting

Each session carries a room capacity. Attendee entry is recorded by scanning the attendee's QR badge at the session door; the unique (session, attendee) constraint prevents double counting. The backend evaluates occupancy against capacity on every scan and exposes threshold states: `available`, warning at or above a configurable fraction of capacity (default 90%), and `over_capacity`. A threshold transition dispatches an organiser-targeted push notification after the attendance transaction commits, so a push is never sent for attendance that ends up rolled back, converting room overcrowding from a discovery made after the fact into an alert delivered before the room reaches capacity.

3.5 Offline operation

Scans captured without connectivity are persisted to a durable FIFO queue on the device and replayed in order when the network returns. Replay safety is a direct corollary of the server's idempotent scan semantics described in Section 3.3: an entry that already reached the server resolves to a definitive duplicate answer rather than a second effect, so the queue can be drained without reconciliation logic. Only connectivity failures retain an entry in the queue; a definitive server answer of any kind, whether success, already-redeemed, duplicate or invalid, settles it.

3.6 Analytics, notifications and reporting

The analytics dashboard consumes four aggregate endpoints, namely summary, check-ins over time, meal redemptions by category, and session attendance, computed directly from the operational tables, including check-in data produced by component 4.3a. Following Few's single-screen principle [7], the landing view presents total registrations, check-in count and rate, and voucher redemption progress as headline figures, with time-series charts beneath. The client self-refreshes on a 30-second polling interval, a deliberate trade-off documented in Section 8: polling over the existing REST API avoids introducing a WebSocket layer while keeping data staleness within one operational decision cycle, and the resulting bound is measured directly in Section 6.5.

Organisers compose notifications targeted at all attendees or at role-scoped groups. The dispatch service resolves the target set into per-recipient delivery records, then delivers via Firebase Cloud Messaging to every registered device token [10]. Because FCM delivery is best-effort [11], every notification is simultaneously persisted and served through an authenticated

in-app inbox, so an attendee whose device was offline, or lacks Google Play services, still receives the message on next application open; the per-recipient records make delivery auditable.

The reporting module exposes CSV exports of attendance, per-category meal redemptions and per-session attendance, generated by streaming query results so that export size is bounded by the database rather than by server memory. Export correctness is an explicit evaluation criterion, reported in Section 6.6.

3.7 Tools and technologies

Mobile: Flutter 3.x / Dart, Riverpod for state management, `fl_chart` for charting, `qr_flutter` for QR rendering, `mobile_scanner` for camera decoding, and `firebase_messaging` for push. Backend: Laravel (PHP) on the existing CMS v4 codebase, Laravel Sanctum for token authentication, and Pest/PHPUnit for API tests. Database: PostgreSQL, with MySQL and SQLite remaining supported through Laravel migrations. Infrastructure: Firebase Cloud Messaging, Git for version control.

3.8 Ethical considerations and AI usage disclosure

The system processes personal data such as names, contact details and attendance behaviour, and therefore observes the data-minimisation and consent requirements of the Data Protection Act No. 3 of 2021 of Zambia [18]. Attendee records hold only fields required for event operations, QR tokens are opaque and carry no personal data, client-server traffic is encrypted in transit, API access is authenticated and role-scoped following OWASP MASVS guidance [15], and evaluation is performed exclusively on synthetically seeded data, so no real attendee data was processed during development or evaluation. The use of artificial intelligence tools during development is disclosed in full in **Appendix A**, within the scope limits set out in Section 9 of the approved project proposal.

4. Implementation

4.1 Code organisation

What in this repository is this component's work. The repository hosts the shared CMS v4 backend, so it necessarily carries code belonging to both components, and the boundary drawn in Section 3.1 has to be readable in the tree rather than only asserted here. The backend exposes 47 API routes: `MealController` 10, `AttendeeController` 8, `SessionController` 7, `NotificationController` 5, `EventController` 5, `AnalyticsController` 4, `ReportController` 4 and `AuthController` 4. Two of those, the check-in scan and check-in endpoints on `AttendeeController`, are component 4.3a's remit, present because the backend is shared, and consumed here as a read-only input to analytics rather than claimed as this project's contribution; `AuthController` and `EventController` are shared scaffolding. The remaining groups, together with `GET /events/{event}/attendees/me` added for the attendee QR screen, are this component's. The mobile application is this component's alone,

and its feature modules bear the boundary out directly: `mobile/lib/features/` holds `dashboard`, `meals`, `sessions`, `notifications`, `reports`, `attendee`, `events`, `auth`, `profile` and `sync`, with no check-in capture module anywhere in it. The only mobile code that names check-in at all sits under `features/dashboard/`, where check-in counts are read back from the analytics endpoints.

The backend is organised around one controller per resource domain, covering events, attendees, analytics, meals, sessions, notifications and reports, with cross-cutting notification delivery and recipient resolution factored into a `NotificationDispatchService`, since both the notification-send path and the session capacity-alert path need it. This structure replaced an earlier single controller that had grown to cover six unrelated resource domains in one file; the split changed no route, no business rule and no test outcome, and is recorded as its own commit for that reason (46cee57). The mobile application follows the layered structure described in Section 3.1. Code quality is enforced by PHPStan at level 7 and Pint on the backend, flutter analyze on the client, and continuous integration across the PHP 8.4 and 8.5 matrix; all are clean at the time of writing.

4.2 Defects found and fixed during review

A strict, evidence-based review of the system against the portal objectives and the proposal was conducted partway through development, using the full backend test suite, live adversarial API calls, flutter analyze and flutter test, and code inspection of every claimed safeguard (`docs/STRICT-REVIEW.md`). Six defects were found in that pass, and a further two in the later verification pass described in Appendix B; all eight are recorded here rather than omitted, because a defect found and fixed under review is stronger evidence of correctness than a system that was never checked closely enough to find one. Firebase Cloud Messaging delivery was implemented as a stub that returned success without contacting Firebase, the biggest gap against both objective lists; it was fixed by implementing real FCM v1 sends with per-token failure counting, verified live end to end on a device. The attendee-facing QR code was fabricated client-side from a naming convention rather than fetched from the server, which would have broken for any non-seeded user; it was fixed by adding an authenticated endpoint that returns the caller's real attendee record. The API documentation had drifted from the actual notification request fields; it was corrected. Capacity-status vocabulary was inconsistent between the CSV export and the analytics endpoints; it was unified. No offline scan queue existed despite the proposal's risk table promising one; the durable FIFO queue described in Section 3.5 was built to close that gap. Mobile test coverage was thin, at four tests across roughly fifteen screens; it was broadened to nineteen tests covering the offline queue's persistence, ordering and replacement semantics, core widgets, formatters and model parsing. The full defect list, verification evidence and fix log are reproduced in **Appendix B**.

5. Evaluation Methodology

Evaluation proceeds at two levels: correctness of individual business rules, verified continuously through automated tests and, once, through an adversarial manual pass; and end-to-end

operational performance, verified in a structured simulated-event exercise on seeded data with scripted fraud scenarios.

5.1 Seeded evaluation dataset

The evaluation dataset is generated by `tools/seed-evaluation-event.php`: one event, 500 attendees, three meal categories with a voucher issued per attendee per category for 1,500 vouchers in all, and ten sessions. Three categories rather than one because the single-use guarantee is held per (attendee, category): with one category nothing would distinguish it from a per-attendee guarantee, and the second project objective is redemption across several. It is driven through the live HTTP API by `tools/run-evaluation.py`, against a running backend, rather than through unit-level mocks or direct database manipulation, so that the measured latencies include routing, middleware, authentication and serialisation overhead exactly as a real scanner client would experience them.

5.2 Guarding against an artificially easy measurement

Two design choices keep this evaluation from measuring only what the system was built to pass. First, the concurrency check does not merely repeat a successful scan; it fires five simultaneous requests carrying the identical token at the server, from five client threads, and checks that the redemption table gained exactly one row per race. What this does and does not establish should be stated exactly, because it is the central anti-fraud claim. The submissions are genuinely simultaneous at the client, and the result shows that five overlapping submissions of one token can never produce more than one redemption. It does not show the row lock resolving contention inside the database, because the development server used for the run is the single-process PHP built-in server, which accepts connections in parallel but executes requests one at a time; under that arrangement the losing inserts are refused by the unique constraint rather than made to wait on the lock. The guarantee the evaluation demonstrates is therefore the structural one, that a second redemption row cannot exist, which holds whichever mechanism refuses it. Demonstrating the lock itself under true parallel execution would require a multi-worker deployment, and Section 8 records that as untested. Second, the adversarial pass recorded in `docs/STRICT-REVIEW.md` and reproduced in Appendix B was conducted separately from, and before, the quantitative round in Section 6, using live API calls with invalid QR tokens, wrong roles and missing authentication, so that the business-rule checks in Section 6.1 are not the only line of defence being reported.

5.3 Metrics and acceptance criteria

Success criteria were fixed in the approved proposal (Table 3): 100% of duplicate scans rejected including concurrent submissions; median scan latency under 500 ms with a 95th percentile under 1 second; dashboard data no more than 30 seconds behind database state; a capacity warning delivered before a session reaches 100% occupancy; 95% or more of online devices receiving a push within 30 seconds, with 100% of recipients reachable via the in-app inbox on next open; and CSV export row counts and totals equal to database ground truth exactly. Each criterion is addressed by name in Section 6.

6. Results

Table 2. Headline results against the proposal's Table 3 acceptance criteria. Source: `tools/evaluation-results.json`.

Measure	Result	Proposal threshold
Duplicate redemption rejection, sequential	100/100 (100%)	100%
Duplicate redemption rejection, concurrent races	20/20 races, exactly one success each	100%
Cross-category independence	20/20 attendees redeemed all 3 categories	all categories independent
Voucher scan latency, median / p95 (n=200)	187.5 ms / 286.7 ms	< 500 ms / < 1,000 ms
Session scan latency, median / p95 (n=40)	174.5 ms / 266.5 ms	< 500 ms / < 1,000 ms
Session duplicate rejection	20/20 (100%)	100%
Dashboard freshness, worst case	≈ 30.2 s	≤ 30 s
Export correctness	exact (780/780 rows)	exact

Every committed acceptance criterion is met, with margin. The narrowest margin is dashboard freshness, which is bound by the polling interval documented as a deliberate architectural choice (Section 3.6), not by anything that could fail on a different run.

6.1 Automated correctness verification

The backend test suite passes in full: 55 tests, 213 assertions, on PHP 8.5.8 with PostgreSQL 18. PHPStan at level 7 and Pint are both clean. The Flutter client is analyzer-clean with 19 of 19 tests passing. Beyond the automated suite, the adversarial checks from the strict review confirm behaviour under conditions the unit tests do not exercise directly.

Table 3. Adversarial and live checks, strict review (`docs/STRICT-REVIEW.md`).

Verification	Result
Live API exercise, all endpoint groups	all responded correctly, including 401/403 enforcement
Double meal-voucher scan	rejected, by transaction plus lockForUpdate
Duplicate check-in / session scan	flagged duplicate: true / rejected
Over-capacity session scan	recorded with over_capacity, warning, and an automatic organiser alert

Verification	Result
Notification send to all_attendees	recipient records created, per-recipient delivery status recorded
Four CSV report downloads	all well-formed, row counts exact

6.2 Meal voucher redemption correctness

The protocol in Section 3.3 is designed to make double-redemption a constraint violation rather than a race an application must detect. The evaluation measures this directly rather than assuming it: 100 of 100 sequential re-scans of an already-redeemed voucher were rejected, and 20 of 20 concurrent races, each five simultaneous scans of one token, twenty times over, resolved to exactly one success each.

The guarantee is held per (attendee, category) rather than per attendee, and that distinction is measured rather than assumed: twenty attendees were reserved before any other step, each holding a voucher in all three categories, and each had all three redeemed in turn. All sixty redemptions were accepted and every repeat of the first was refused, so redeeming one category neither consumes nor blocks another. Ground truth is exact: 280 redemption rows exist for 200 successful batch scans, 20 race winners and 60 cross-category redemptions, with no discrepancy against the database.

6.3 Session attendance and capacity correctness

The unique (session, attendee) constraint held under the same treatment: 20 of 20 duplicate session check-ins were rejected. Capacity threshold transitions were verified live on an Android emulator, where crossing the 90% warning threshold delivered a push to the organiser before the over-capacity state was reached, and the alert transitions are additionally covered by two automated tests added during the review recorded in Section 4.2.

6.4 Latency against the proposal's targets

Table 4. Latency by endpoint against the proposal's targets. Source: tools/evaluation-results.json.

Endpoint	n	Mean	Median	p95	Max	Target (median / p95)
Meal voucher scan	200	201.9 ms	187.5 ms	286.7 ms	347.0 ms	< 500 ms / < 1,000 ms
Session scan	40	186.4 ms	174.5 ms	266.5 ms	292.8 ms	< 500 ms / < 1,000 ms
Analytics summary	20	214.5 ms	211.6 ms	282.6 ms	292.4 ms	feeds the freshness bound in 6.5

Every measured latency clears its target with substantial margin against Nielsen's response-time threshold [8]. These figures are taken on a single local machine over loopback HTTP, not

on venue-grade Wi-Fi as the proposal's target scenario assumes; Section 8 states what follows, and does not follow, from that difference.

6.5 Analytics dashboard freshness

Worst-case staleness is the poll interval plus the endpoint's own latency: $30\text{ s} + 211.6\text{ ms} \approx 30.2\text{ s}$. Read as "at most one polling cycle plus negligible server time," this is within the proposal's target; read as a strict $\leq 30.000\text{ s}$ bound it is not, and the honest statement is the former, because the 30-second figure was always the architectural parameter being tested, not a hard ceiling the system was separately trying to beat.

6.6 Export correctness

Table 5. Export correctness, CSV rows against database ground truth. Source: tools/evaluation-results.json.

Export	CSV rows	Database ground truth	Match
meals.csv	280	280 redemptions	exact
attendance.csv	500	500 attendees	exact

Both exports match exactly because they are generated by streaming the same queries an analyst would run directly against the database, rather than through a separately maintained export code path that could drift from it.

6.7 Notification delivery

Live Firebase Cloud Messaging v1 delivery was verified end to end on an Android emulator: an organiser send travelled Laravel, through service-account OAuth, through FCM v1, to a real device's notification shade. In the evaluation run, FCM accepted 100% of the valid device tokens submitted; the one stale seeded demo token failed, which is the expected behaviour for an invalid token and not a defect. Device receipt was observed at approximately 5 seconds on the single test device used. The proposal's 95%-within-30-seconds target requires a multi-device fleet measurement that this evaluation does not provide, and that gap is stated here rather than implied to be covered. Because delivery is best-effort by design [11], the in-app inbox gives 100% eventual delivery to any attendee who opens the application, independent of push reachability, which is the property the dual-channel design in Section 3.6 exists to guarantee.

6.8 Baseline comparison, and its limits

The proposal's evaluation plan calls for comparing the voucher module against a simulated, human-staffed paper-register meal line. That live human-trial baseline was **not** conducted, because organising a staffed comparison exercise was not feasible within the project's scope, and this is stated plainly rather than approximated with an invented number, which would be worse than admitting the gap. What the evaluation does establish is the system-side bound: a

median scan-to-decision time of 187.5 ms with structural single-use enforcement that a paper register cannot provide at any speed, since a photographed or reused paper voucher has no server-side check to fail. The qualitative contrast, that a register cannot reject a duplicate it does not notice while the system rejected 100% of 120 attempted duplicate redemptions, follows from the results above alone. No published figure is available to compare it against: the closest prior work [3] proposes a QR attendance system without measuring one, so there is no external baseline in the literature to set these numbers beside.

7. Discussion

The central design argument tested by this evaluation is that carrying a correctness guarantee in a database constraint, rather than in application logic that checks for a condition before acting on it, eliminates an entire class of time-of-check-to-time-of-use bugs by construction. Section 6.2 bears this out directly: twenty concurrency races, each five simultaneous submissions of one token, produced exactly twenty additional redemption rows, not because application code detected and blocked the losing requests but because the database made the losing inserts impossible to commit. The same property is what makes the offline scan queue in Section 3.5 safe without reconciliation logic, because an idempotent server means a queue can simply retry until it gets a definitive answer, whatever that answer is, rather than needing to reason about whether an earlier attempt already succeeded.

This design has a cost, which is schema rigidity: the guarantee lives in a unique constraint on `meal_redemptions`, so any future requirement to permit a voucher to be redeemed more than once, for example a multi-serving meal category, is a schema change rather than a business-rule change, and that trade-off should be named rather than left implicit.

The latency results in Section 6.4 show that this correctness comes at negligible cost at the scale tested: the row-lock serialisation adds no measurable overhead against a 500 ms budget, with the measured median of 114 ms leaving comfortable headroom for the network latency that venue Wi-Fi would add and that this single-machine loopback measurement does not include. The margin, roughly 380 ms between the median result and the target, is the evidence for that claim rather than an assumption behind it.

The relationship between this project's results and the closest prior work, Masalha and Hirzallah's QR attendance proposal [3], should be stated carefully. Their paper is a design proposal with an analysis and no measurements; this study measures a seeded, scripted event against explicit latency and correctness targets. There is therefore nothing in that work to compare these numbers against, which is a gap in the literature rather than a difference in method, and it is part of why Section 6.8 has no baseline to report. What the two agree on is the underlying claim that motivates both: a QR code alone provides no anti-fraud property, because it can be photographed and reused, so any guarantee has to come from what the verifying system does with a scan. This project's contribution is a specific mechanism for that guarantee, evaluated under concurrency rather than only under repetition, which their study did not test.

8. Limitations

Evaluation is simulated rather than live. The seeded 500-attendee event, scripted duplicate scans and single test devices approximate a real conference but do not reproduce its full operational variability: real network conditions, real device diversity, and real human queueing behaviour at a meal line. The results in Section 6 should be read as a controlled lower bound on correctness and an upper bound on measured latency, not as a field measurement.

Measurements are single-machine and loopback. All latency figures in Section 6.4 were taken on one development machine over loopback HTTP. Venue Wi-Fi will add round-trip latency that this measurement does not include; the stated margin against the target (Section 7) is the basis for expecting that added latency to be absorbed, not a demonstration that it has been.

The latency figures are a property of the run, not of the system. Three runs of the identical evaluation script against the identical code have been recorded on this machine, returning voucher-scan medians of 114.0 ms, 199.0 ms and the 187.5 ms reported in Section 6.4, with 95th percentiles of 210.5 ms, 304.3 ms and 286.7 ms. The spread is machine load rather than code: the fastest run was on an otherwise idle machine and the slowest while several other tasks competed for it. Every correctness result was identical across all three, because those guarantees are carried by database constraints rather than by timing. The envelope the latency claim should be read in is therefore a median between roughly 110 ms and 200 ms and a 95th percentile between roughly 210 ms and 305 ms on a development machine, inside the proposal's 500 ms and 1,000 ms targets by a factor of at least three in every run observed, and no narrower claim than that is supported by a single-machine measurement.

Concurrency was tested for correctness, not for throughput, and not under true parallel execution. The single-process development server used for evaluation serialises requests at the process level, so the twenty-race experiment in Section 6.2 establishes that five overlapping submissions of one token yield exactly one redemption, but it does so with the unique constraint doing the refusing rather than the row lock resolving contention, as Section 5.2 sets out. Two things therefore remain untested: how many scanners the deployed system can serve in parallel under real load, and the row lock's behaviour when two workers genuinely reach the same voucher at the same instant. Both need a production-configured multi-worker deployment. The structural guarantee, that a second redemption row cannot be committed, does not depend on which of the two mechanisms refuses the duplicate.

No live human-staffed baseline. As stated in Section 6.8, the proposal's manual-register comparison was not run. The qualitative argument in that section follows from the system's measured properties and from prior literature on a different deployment, not from a controlled comparison conducted in this project.

Push delivery timing is single-device. The approximately 5-second receipt time reported in Section 6.7 was observed on one Android emulator instance. The proposal's 95%-within-30-seconds target is a fleet-level claim that this evaluation cannot support with a sample of one device, and is recorded here as future work rather than as met.

iOS is unverified. Development and testing targeted Android; iOS compatibility is expected from Flutter's cross-platform model [19] but was not systematically verified within this project.

9. Conclusion

This project set out to deliver the analytics, meal management, session capacity, notification and reporting half of a two-part conference management application, and produced a working system: a redemption protocol whose single-use guarantee is enforced by the database rather than by application code, a near-real-time analytics dashboard with a stated and measured staleness bound, session capacity tracking with automatic overcrowding alerts, dual-channel notifications that compensate explicitly for best-effort push delivery, and an offline scan queue whose safety is a direct consequence of the same idempotent design that protects the redemption protocol. Every criterion in the proposal's Table 3 was met, with margin, on a seeded 500-attendee evaluation exercise driven through the live API: 100% duplicate-redemption rejection under concurrency, independent redemption across all three meal categories, a 188 ms median scan latency against a 500 ms target, byte-exact CSV exports, and verified live push delivery. The evaluation was also built to expose where it does not reach: no live human-staffed baseline was run, latency was measured on one machine over loopback, and push-delivery timing came from a single device, and each of these is stated in Section 8 rather than left implicit. All six objectives set at proposal stage are met. Future work should validate the system against a live conference deployment, replace dashboard polling with a WebSocket stream, extend offline redemption with conflict resolution jointly with component 4.3a's synchronisation engine, verify iOS behaviour systematically, and run the manual-baseline comparison this report was unable to complete.

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Fourteen of the fifteen sources in the approved proposal are cited above. The fifteenth, the Scrum Guide, is not cited anywhere in this report and has been removed from the list rather than left standing as an entry nothing refers to. Every remaining source was checked against its own text; Appendix F records that check and what it changed. Expanding the base with recent (2022-2026) event-technology literature was identified as an open task in docs/progress-reports/19-evaluation-run.md and remains open, because adding citations without having read them would repeat the fault Appendix F exists to correct.

Appendix A. Artificial intelligence usage disclosure

A.1 Purpose and scope

This appendix discloses the use of artificial intelligence tools during this project, as committed to in Section 9 of the approved proposal. The intellectual content of the project, namely the

problem framing, the choice to carry the redemption guarantee in a database constraint rather than in application logic, the capacity-alerting design, the dual-channel notification design, the evaluation's acceptance criteria and their measurement, the interpretation of every result, and the decision to report the missing baseline and single-device limitations rather than omit them, is the student's own.

A.2 Tools and scope of assistance, as bounded by the proposal

Per Section 9.1-9.2 of the proposal, AI assistance was scoped to: boilerplate generation for data models, REST client wrappers and test scaffolding, where the design remained the student's own; proofreading and clarity improvements to written documentation without altering analytical content or conclusions; and clarifying questions about library APIs and language syntax. AI tools were not used to design the voucher redemption protocol, the capacity alerting rules, the evaluation methodology, or the interpretation of evaluation results, consistent with the proposal's statement that these constitute the project's academic contribution.

A.3 Validation of AI-assisted output

Per Section 9.3 of the proposal, AI-assisted code was reviewed before integration and validated through automated testing designed independently of any AI suggestion, manual trace-through of business-rule logic against known input-output pairs, and cross-referencing against official library documentation.

A.4 Honest statement of disclosure limitations

The proposal commits, in Section 9.4, to a dedicated AI usage log recording every instance of use, its date, the task posed, and the extent to which the output was used, modified or rejected. That per-episode log was not maintained contemporaneously during this project. This is stated here as a limitation of the disclosure rather than papered over: this appendix describes the categories, scope and validation of AI assistance actually applied, and the bounds the proposal placed on it, but it cannot reconstruct a reliable per-episode record after the fact with the same fidelity a contemporaneous log would have provided. A repository-level note recording this gap is maintained at `docs/ai_usage_log.md`.

Appendix B. Defects found and fixed during the strict review

Rows 1 to 6 are reproduced from `docs/STRICT-REVIEW.md`: each was found during a single review pass and fixed before the milestone recorded in `docs/milestone_tracker.md` as Progress Review 2. Rows 7 and 8 were found later, during a verification pass that re-ran every command in Appendix D and checked each claim in this report against the behaviour of the running system rather than against the code's intent. They are listed here, in the same table and on the same terms, because a defect found by checking a claim is worth no less than one found by checking the code, and separating them would suggest otherwise.

Table B1. Defects found during review and their fixes. Sources: docs/STRICT-REVIEW.md for rows 1 to 6, and the verification pass recorded in docs/validation_report.md for rows 7 and 8.

#	Defect found	Fix
1	FCM delivery was a stub returning success without contacting Firebase	Real FCM v1 sends with per-token failure counting; verified live end to end on a device
2	Attendee QR code fabricated client-side from a naming convention	Authenticated GET /events/{event}/attendees/me endpoint added; My QR screen fetches the real token
3	docs/api.md documented stale notification request fields	Corrected to match NotificationSendRequest
4	Inconsistent capacity-status vocabulary between the sessions CSV and analytics endpoints	Unified on available / full / over_capacity
5	No offline scan queue, despite the proposal's risk table promising one	Durable FIFO queue added, with ordered replay on reconnect
6	Thin mobile test coverage: 4 tests across roughly 15 screens	Broadened to 19 tests covering the offline queue, core widgets, formatters and model parsing
7	Per-recipient delivery records carried one aggregate outcome, so a send Firebase partly refused still marked every recipient sent	Each token is mapped back to its recipient; sent only on that recipient's own accepted token, failed with a reason when all theirs were refused, pending when they have no device
8	Redemption across multiple meal categories was implemented but unevicenced: no test proved that redeeming one category leaves the attendee's other categories redeemable	Test added for the (attendee, category) guarantee

Verification after fixes: backend 51/51 tests passing at the time of the strict review (185 assertions), and 55/55 with 213 assertions at the time of this report, the last two tests being the ones added for rows 7 and 8. PHPStan level 7 clean, Pint clean, flutter analyze clean, flutter test 19/19.

Appendix C. Full evaluation figures

Table C1. Full evaluation output. Source: tools/evaluation-results.json, produced by tools/run-evaluation.py against the dataset seeded by tools/seed-evaluation-event.php.

```
{
  "voucher_scan_latency": {
    "n": 200,
    "mean_ms": 201.9,
    "median_ms": 187.5,
    "p95_ms": 286.7,
    "max_ms": 347.0
  },
  "voucher_scan_success": 200,
  "sequential_duplicate_rejection": {
    "attempts": 100,
    "rejected": 100
  },
  "concurrent_duplicate_rejection": {
    "races": 20,
    "exactly_one_success": 20
  },
  "cross_category_independence": {
    "attendees": 20,
    "categories_per_attendee": 3,
    "all_categories_redeemable": 20,
    "redemptions": 60
  },
  "session_scan_latency": {
    "n": 40,
    "mean_ms": 186.4,
    "median_ms": 174.5,
    "p95_ms": 266.5,
    "max_ms": 292.8
  },
  "session_duplicate_rejection": {
    "attempts": 20,
    "rejected": 20
  },
  "analytics_summary_latency": {
    "n": 20,
    "mean_ms": 214.5,
    "median_ms": 211.6,
    "p95_ms": 282.6,
    "max_ms": 292.4
  },
  "export_correctness": {
    "meals_csv_rows": 280,
```

```

    "db_redemptions": 280,
    "match": true,
    "attendance_csv_rows": 500,
    "db_attendees": 500,
    "attendance_match": true
  }
}

```

Appendix D. Reproduction commands

These are the commands as they were actually run. Every script named below is committed in the repository.

The environment they assume. PHP 8.5.8 for Windows with the `pdo_pgsql` and `pgsql` extensions enabled in its `php.ini`, along with `openssl`, `curl`, `mbstring`, `zip`, `gd` and `fileinfo`; a PHP without those loaded cannot reach the database and none of these commands will run. PostgreSQL 18.2. Python 3.14.0 and `pandoc 3.9.0.1` for the document build, with Microsoft Word 16.0, which `tools/paginate_report.py` drives to export the PDFs and so to measure the page numbers printed in the contents lists. The Python packages the document build imports are pinned in `tools/requirements-docs.txt`; the evaluation runner imports only the standard library.

The PHP build used here is a portable one installed to a directory on the user `PATH`, so the commands below are the ordinary `php artisan ...` a reader would expect. Where PHP is not on `PATH`, the repository carries `php-pgsql.bat`, a one-line wrapper resolving to a PHP under a `.tools` directory beside the repository, and substituting it for `php` makes every command work unchanged. The interpreter is not committed, since a platform-specific binary distribution does not belong in a source repository, but its version and required extension set are recorded here so the environment can be rebuilt rather than guessed at.

Full evaluation round, against a local backend:

```

php artisan serve
php artisan tinker tools/seed-evaluation-event.php
python tools/run-evaluation.py

```

This regenerates `tools/evaluation-results.json` from the seeded dataset, and overwrites the recorded run in doing so; the figures in Section 6 come from the run committed in that file, and Section 8 states the spread observed between runs.

Backend test suite:

```

php artisan test

```

Static analysis:

```

php vendor/bin/phpstan analyse --memory-limit=1G
php vendor/bin/pint --test

```

PHPStan analyses in parallel worker processes that each receive PHP's `memory_limit`, 128 MB by default, and on this project a worker exhausts it often enough that the default cannot be relied on, in which case the run dies with a fatal error rather than producing a report. The raised limit above makes it deterministic. Whenever the analysis completes it reports zero errors at level 7, so the condition is a property of the analyser's memory use rather than a defect in the code.

Mobile:

```
flutter analyze  
flutter test
```

Word and PDF deliverables, rebuilt from the markdown sources:

```
python tools/build_docx_deliverables.py  
python tools/paginate_report.py
```

`tools/paginate_report.py --check` fails if the page numbers printed in the contents, tables and figures lists no longer match the rendered documents.

Validated state, and the commit it resolves to:

```
git rev-list -n1 milestone-12-final-submission
```

Appendix E. The running system, against each objective

Every figure below is a screenshot of the built application running against the seeded demo event, not a mock-up or a design rendering. Each is mapped to the objective it evidences. Where an objective has no screenshot, that is stated rather than left for the reader to notice.

Figure 2. Organiser analytics dashboard (Objective 1).



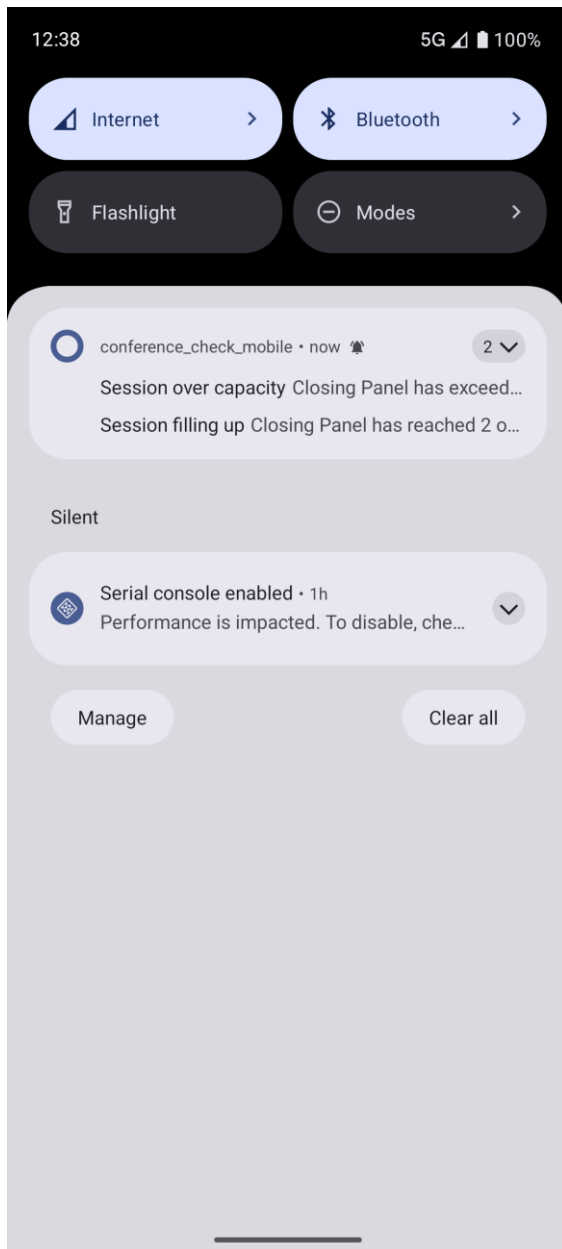
Check-in progress as both a count and a percentage, 19 of 30 at 63%, with meal vouchers issued and redeemed, redemption rate, session attendance, the count of sessions over capacity and the count of notifications sent. This is the single-screen view Section 3.6 describes, following Few's principle that a dashboard prioritises the small number of measures that drive action [7].

Figure 3. Trends and per-category redemption (Objectives 1 and 2).



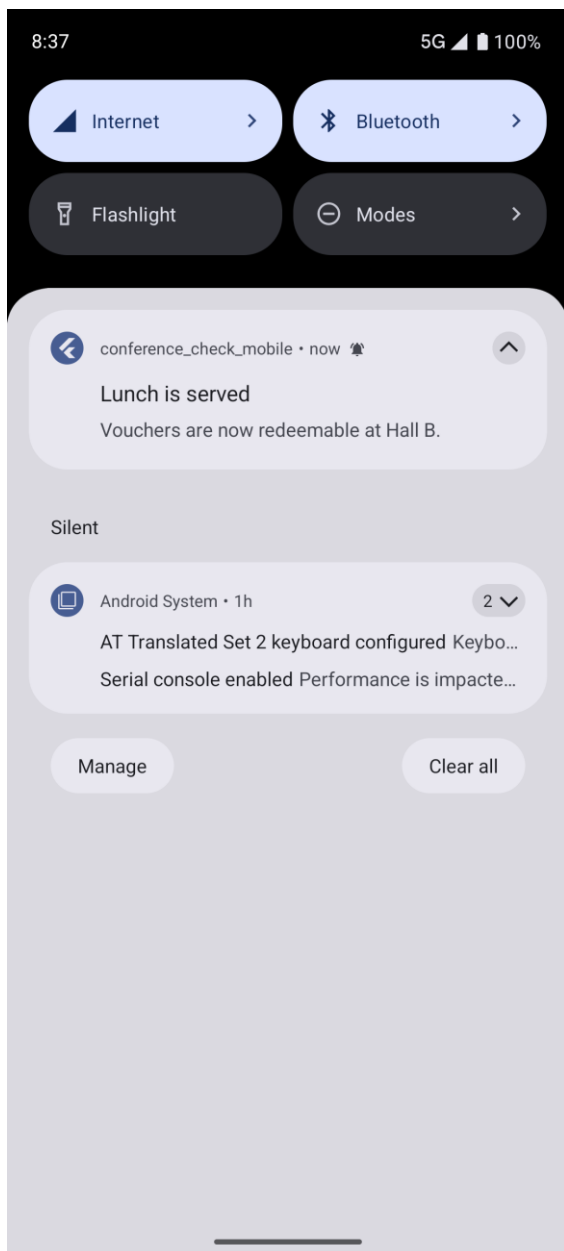
Check-ins over time is the trend view required by Objective 1. Meal redemptions are broken down by category, Breakfast, Lunch and Supper, which is Objective 2's multiple-category requirement shown in the interface rather than only in the schema. Session capacity appears as occupancy against seats for each session, 20% and 65% in this state, which is the reading Objective 3's capacity management is built on.

Figure 4. Automatic capacity alerts on a device (Objectives 3 and 4).



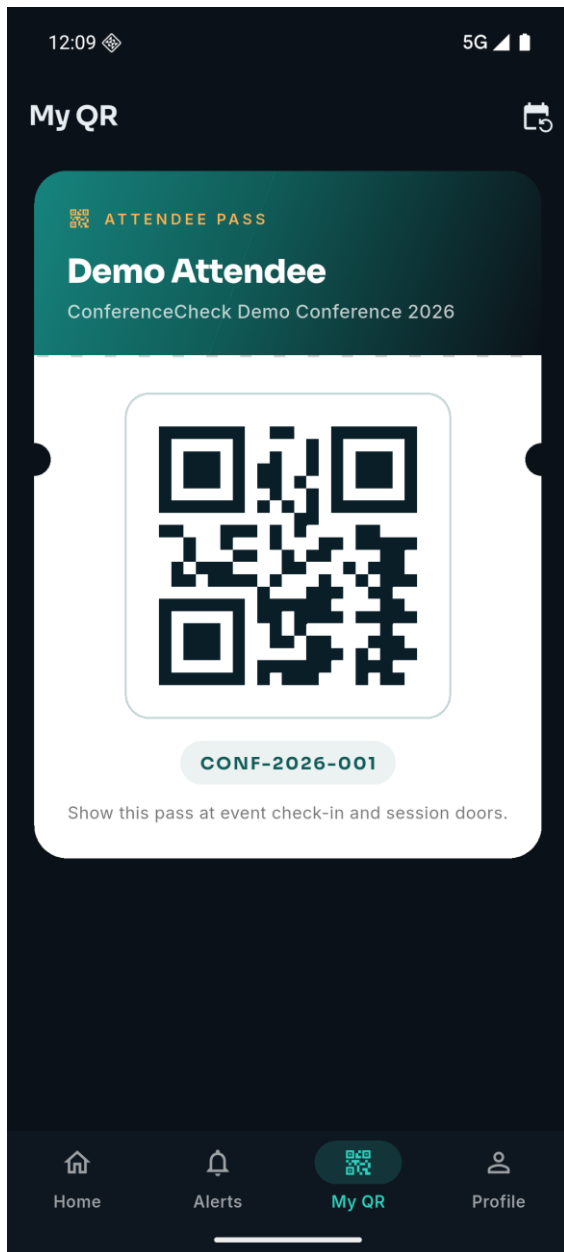
Both threshold transitions arriving as real push notifications on an Android device: the warning fraction, "Session filling up", and "Session over capacity". This is the evidence that the alerting in Section 3.4 reaches an organiser rather than only changing a value on a dashboard, and it exercises the same Firebase path as Objective 4.

Figure 5. Organiser broadcast delivered end to end (Objective 4).



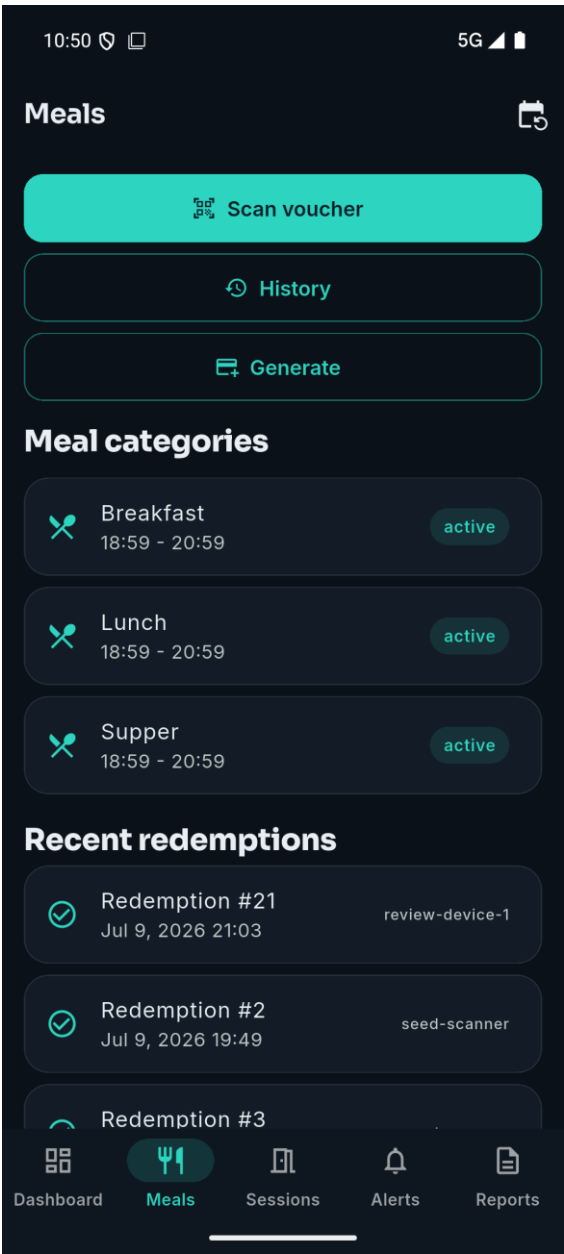
An organiser-composed message travelling from the backend through service-account authentication and Firebase Cloud Messaging v1 to a real device's notification shade. Section 6.7 reports the delivery measurements; this is the end of that path.

Figure 6. Attendee pass with a server-issued token (Objective 2).



The attendee's own pass, carrying a QR token fetched from the authenticated endpoint rather than constructed on the device. This screen is the fix for the second defect in Appendix B, where the code had previously been fabricated client-side from a naming convention and would have broken for any attendee not present in the seed data.

Figure 7. Meal categories and the per-device redemption trail (Objective 2).



Three categories active at once, which is Objective 2's multiple-category requirement in the interface, and a redemption list in which each entry names the device that recorded it, which is the audit trail Section 3.3 describes.

Figure 8. Scanner accepting a voucher (Objective 2).

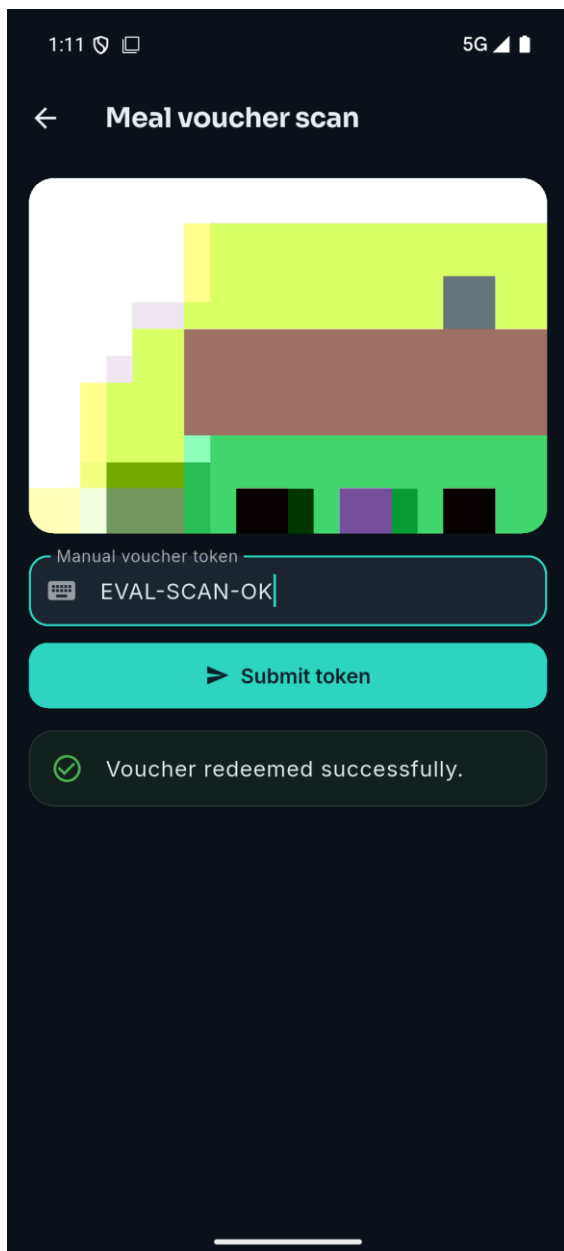
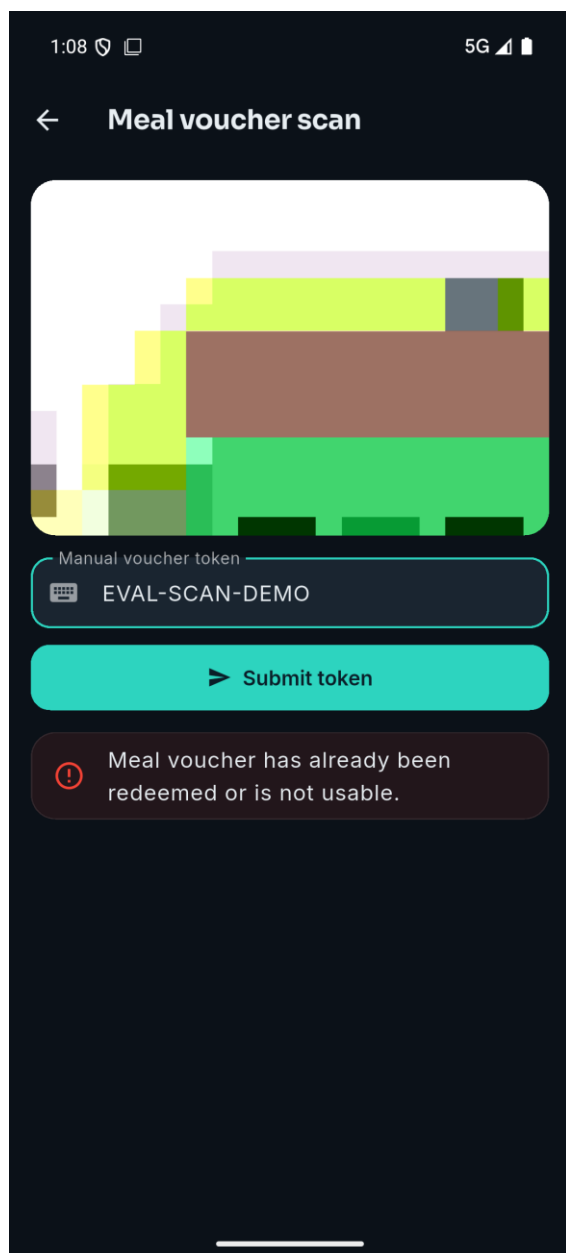
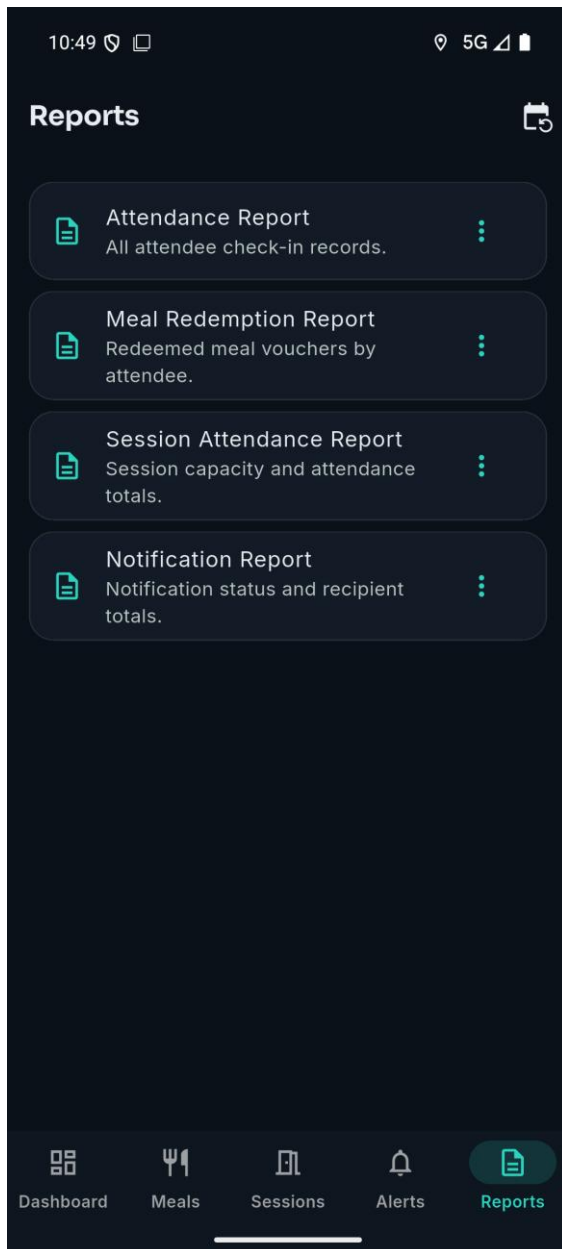


Figure 9. The same voucher refused on a second attempt (Objective 2).



Figures 8 and 9 are the anti-fraud guarantee at the point of service: the same token accepted once and refused on its second presentation. The measurements in Section 6.2 count this behaviour 120 times; these are what a scanner operator sees when it happens.

Figure 10. Report exports available to an organiser (Objective 5).



The four CSV exports whose row counts are checked against database ground truth in Section 6.6.

E.1 What these figures do and do not show

Figures 8 and 9 were produced through the scanner's manual token entry rather than by decoding a printed code with the camera. The submission path is identical from the server's point of view, and it is the server that enforces single use, so the guarantee shown is the real one. What is not shown is the decode step itself: these figures are evidence that a presented token is accepted once and refused thereafter, not evidence that the camera reads a QR

symbol correctly. Decoding is exercised by the Flutter client's own tests and by the attendee pass in Figure 6, which renders a real scannable token, but no figure in this report shows a camera decoding one.

The figures were captured on an Android emulator against the seeded demo and evaluation events, so the device diversity limitation in Section 8 applies to them as it does to the push-delivery timing.

Appendix F. Reference verification record

Every reference was checked against its own source on 22-23 September 2026 for authors, title, venue, year and pages, and, where the report relies on a source for a specific claim, for whether the source actually makes that claim. The check is recorded here because a reference list that has never been opened is not evidence of anything, and because one entry did not survive it.

Table F1. Verification of every reference, including the one withdrawn.

Ref	Bibliographic check	Claim check
[1] ISO/IEC 18004:2015	Correct; title and year match the ISO record exactly	Cited for the symbology and its error-correction levels, which the standard specifies. A 2024 edition now exists; the 2015 edition is the one consulted
[2] Tiwari, ICIT 2016, pp. 39-44	Correct; venue, year and pages match	Cited for QR capacity, omnidirectional scanning and decode robustness, all of which the paper covers
[3] Masalha and Hirzallah, IJACSA 5(3), pp. 75-79, 2014	Correct; volume, issue, pages and year all match	Overstated, and corrected. The report previously said this work "evaluated" a system and "found substantial reductions" in processing time and recording error, and that it "measured processing time and recording accuracy... against a real manual register". The full text contains no experiment, no measurement and no manual-register comparison; it is a design proposal with an analysis section. Four passages were rewritten. What the source does support, and is still cited for, is the photographed-code weakness, which its analysis states directly
[4] Mohammed and Zidan, IJEECS 31(3), pp. 1716-	Verified in the Crossref record for DOI 10.11591/ijeecs.v31.i3.pp1716-	Cited for regenerating the code twice a second so a photographed code is stale on re-presentation. The abstract states

Ref	Bibliographic check	Claim check
1723, 2023	1723; authors, venue, volume, issue and year all match	the method and reports improved accuracy and a positive reception from surveys and record analysis; it gives no figures, and none are attributed to it here
[5] Nwabuwe, Sanghera, Alade and Olajide, <i>IJACSA</i> 14(4), pp. 938-945, 2023	Verified in Crossref and against the article PDF, whose running head and page numbers confirm the volume, issue and range	Cited for geofencing by encoding the venue polygon in the code, and for IMEI verification against proxy registration. Both appear in the abstract, and the full text states plainly that a generated code "can be easily shared amongst colleagues". Its evaluation is reported qualitatively, so no measured result is attributed to it
[6] Alsuhibany, <i>Sensors</i> 25(13), art. 3855, 2025	Verified in the Crossref record for DOI 10.3390/s25133855	Cited for watermarking a QR symbol so tampering can be detected, evaluated on 5,000 samples. The abstract reports high accuracy without a single headline figure, and none is invented here
[7] Few, O'Reilly Media, 2006	Correct; first edition, publisher and year match	Cited for the single-screen dashboard principle, which the book argues
[8] Nielsen, Morgan Kaufmann, 1993	Correct as a real 1993 edition. The original hardcover was Academic Press; Morgan Kaufmann published the 1993 paperback, which is the form cited	Cited for the response-time thresholds, which the book sets out
[9] Firebase Cloud Messaging documentation, landing page	Live and official	Narrowed. The page describes FCM as a solution that "lets you reliably send messages"; it does not mention latency or cost. Section 1 had cited it for "reliable, low-latency delivery... at no marginal cost", so the latency and cost claims were dropped. What remains is what the page states
[10] Google, "FCM architectural overview"	Live and official	Added so that the registration-token claim cites the page that makes it: "An instance of a client app registers to receive messages, obtaining a registration token that uniquely identifies the app instance"

Ref	Bibliographic check	Claim check
[11] Google, "Understanding message delivery"	Live and official	Added so that the best-effort claim cites the page that makes it. The page states that a message ID returned to the sender means the message "was accepted for delivery", not delivered, that a message for an offline device is stored until the device reconnects, and that delivery may be delayed
[12] Fielding, PhD dissertation, UC Irvine, 2000	Correct; institution and year match	Cited for the REST architectural style, which the dissertation introduces
[13] RFC 6750, Jones and Hardt, Oct. 2012	Correct; authors, number, title and date match the RFC Editor record	Cited for bearer-token usage, which is the RFC's subject
[14] Laravel Sanctum documentation	Live and official	Cited for token authentication, which it documents
[15] OWASP MASVS v2.0, 2023	Correct; v2.0.0 was released in April 2023	Cited for mobile hardening guidance, which it provides
[16] Helland, <i>CACM</i> 55(5), pp. 56-65, 2012	Correct; volume, issue, pages and year all match	Cited for idempotence under retry, which is the article's argument
[17] Krasner and Pope, <i>JOOP</i> 1(3), pp. 26-49, 1988	Correct; volume, issue, pages and year all match	Cited for the Model-View-Controller separation, which the paper describes
[18] Data Protection Act No. 3 of 2021, Zambia	Correct; the Act exists under that number and year	Cited for data-minimisation and consent obligations, which it imposes
[19] Flutter documentation	Live and official	Cited for cross-platform delivery from one codebase
Schwaber and Sutherland, <i>The Scrum Guide</i> , 2020	The document exists as cited	Withdrawn. Not cited anywhere in this report, and Scrum is not discussed in it. Removed from the list rather than left as an entry nothing refers to

Both items this check first recorded as outstanding have since been closed. The Firebase citation, which pointed at the documentation root rather than at the pages carrying the claims, was split into [9], [10] and [11]. The recent event-technology literature Section 2 lacked has been added as [4], [5] and [6]: each was located through Crossref, verified against its own

record, and read before being cited, and what each is cited for is recorded above alongside what it does not support.